Generative AI and Retrieval Augmented Generation

Generative AI

Generative AI refers to models that can create new content such as text, images, audio, video, or code. Large language models are a major example of generative AI.

Embeddings

An embedding is a numerical representation of data such as text. Similar meanings can be represented by vectors that are close to one another in an embedding space.

Retrieval Augmented Generation

Retrieval Augmented Generation (RAG) combines information retrieval with text generation. A retriever first finds relevant documents, and a language model then uses the retrieved information to generate an answer.

Why Retrieval Matters

Retrieval can provide a language model with information from a specific knowledge base. This can make answers more relevant to the available documents and reduce reliance on information stored only in the model.

RAG Pipeline

A simple RAG pipeline contains document ingestion, document storage or indexing, retrieval, and answer generation. Retrieval quality is important because poor retrieved documents can lead to poor answers.